Instructions for Continued Airworthiness
McFarlane Aviation Inc. FAA-PMA Part Number 1111 Stabilizer Screwjack Actuator Kit.
FAA PMA Number: PQ3732CE

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mbly Instructions:

1. Align the alignment marks on the screw bar flange and the sprocket and insert sprocket over the end of the screw bar to its deepest point. *Note: The alignment marks indicate the best fit of the retainer pin (roll pin).*
2. Install retainer pin and secure the pin with safety wire through the pin center. Caution: Use only included pin. Pin is strength critical.
3. Omit Cessna Part Number 0712500-19 barrel cap.
4. Finish installation and rigging per Cessna Service Manual instructions. Caution: The threaded screw bar, bearing, and sprocket are match drilled as a set and selectively fit with the barrel. Do NOT interchange components.

TROUBLESHOOTING

Refer to the applicable Cessna Model Service Manual for troubleshooting instructions and the applicable Cessna Illustrated Parts Catalog for component part numbers. See listing in Data Section below. For troubleshooting, refer to Figure 11-3 in Manual D138-13, Figure 9-3 in Manual D637-13, and Figure 10-3 in Manual D2000C4-13. The following troubleshooting instructions, specific to the McFarlane parts, apply:

1. If the stabilizer is sluggish or jams, clean the barrel and screw jack and re-apply a coating of MIL-PRF-23827 low temperature grease.

PLACARDS
None applicable

DATA
All information to support the continued airworthiness of this replacement part is as defined herein and contained in:

- Relevant Cessna Model Service Manuals.
  - 100 series Service Manual D138-13 (1962 and Prior)
  - 100 series Service Manual D637-13 (1963 thru 1968)
  - 180 & 185 Service Manual D2000C4-13 (1969 thru 1976)
- Relevant Cessna Model Illustrated Parts Catalog.
  - 180/185/A185 IPC P527-12 (1961-1973) dated 6/15/1974 or P699-12 (1974-1986) dated 2/3/1997
  - 180/182 IPC P259-12 dated 6/1/1967

INSPECTION

RECOMMENDED OVERHAUL PERIODS
No additional overhaul time limitations exist with the use of these parts.

AIRWORTHINESS LIMITATIONS
The Airworthiness Limitations section is FAA approved and specifies maintenance required under Sec. 43.16 and 91.403 of the Federal Aviation Regulations unless an alternative program has been FAA approved. No additional airworthiness limitations exist.

ASSISTANCE & REVISIONS
ICA revisions shall be made available on the McFarlane website, www.mcfarlaneaviation.com/ICA. For questions or assistance regarding these Instructions for Continued Airworthiness (ICA), contact McFarlane Aviation, Inc via email or phone. Email: engineering@mcfarlaneaviation.com Phone: 1-800-544-8594 (within the US) or 1-785-594-2741.

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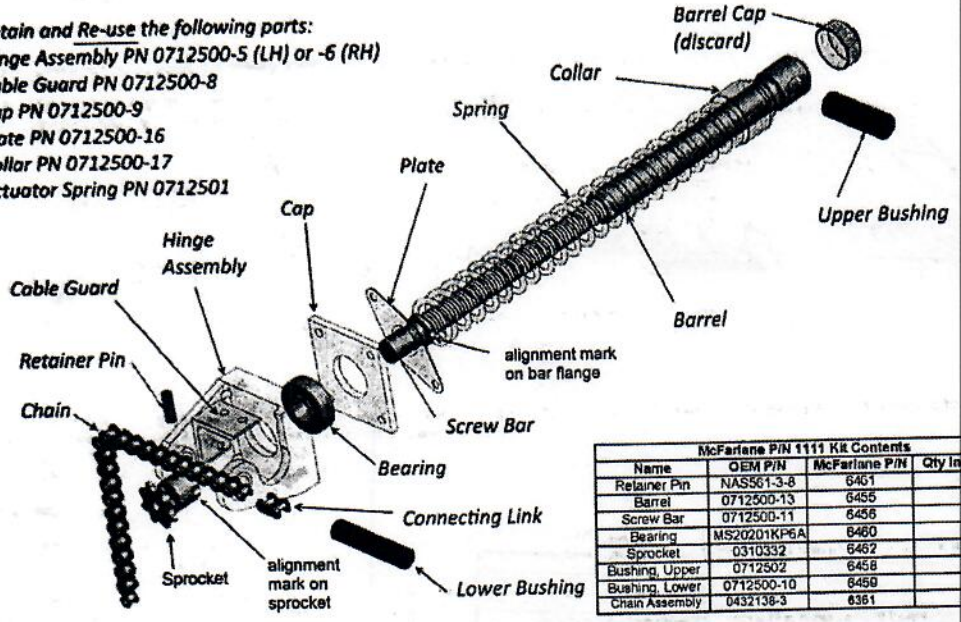
INTRODUCTION
This document is intended to provide for the continued airworthiness of McFarlane Aviation, Inc. PMA Stabilizer Screwjack Actuator Kit eligible for installation on various Cessna model aircraft. The kit part number is 1111 and is comprised of the parts required to overhaul two (2) actuators; (there are two such actuators on each aircraft). This document specifically concerns the installation of the overhaul kit. For all items not related to the installation of the McFarlane Aviation, Inc. overhaul kit, refer to the basic airplane model service and parts manuals.

SYSTEM DESCRIPTION
Cessna Model Series 180, 185, and 182 (through 182D) employ two screw driven actuators to manipulate the position of the horizontal stabilizers. Due to wear and corrosion, overhaul of these actuators is routinely required. Replacement of the sprocket, bushings, bearing, retainer collar, screw bar, barrel, retainer pin, and chain/chain pin may be performed to return the part to optimal condition.

Retain and Re-use the following parts:
Hinge Assembly PN 0712500-5 (LH) or -6 (RH)
Cable Guard PN 0712500-8
Cap PN 0712500-9
Plate PN 0712500-16
Collar PN 0712500-17
Actuator Spring PN 0712501



McFarlane P/N 1111 Kit Contents			
Name	OEM P/N	McFarlane P/N	Qty Included
Retainer Pin	NAS561-3-8	6461	2
Barrel	0712500-13	6455	2
Screw Bar	0712500-11	6456	2
Bearing	MS20201KP6A	6460	2
Sprocket	0310332	6462	2
Bushing, Upper	0712502	6458	2
Bushing, Lower	0712500-10	6459	2
Chain Assembly	0432138-3	6361	1

Note: McFarlane P/N 6457 is an assembly consisting of part numbers 6456, 6460, and 6462, which have been match drilled as a set. P/N 6361 is an assembly consisting of a length of # 25 chain and two connecting links.

The McFarlane Aviation, Inc. overhaul kit provides the parts indicated. Cessna Aircraft does not provide these parts in a kit form. The McFarlane PMA barrel, screw bar, bushings, and sprocket differ from the OEM parts in that they are made of corrosion resistant alloys (duplex stainless steel or Monel); providing for superior corrosion resistance and longer part life.

Note: The McFarlane Sprocket replaces both the original sprocket (Cessna P/N 0310332) and the retaining collar (Cessna P/N 0712500-7).

SPECIAL OPERATING INFORMATION
The control and operation of the actuator does not change with the installation of the overhaul kit; see applicable Cessna Model Service Manual for the operational control of this system.

PART REMOVAL, REPLACEMENT, AND SERVICE INFORMATION
Remove the Cessna stabilizer actuators and install the McFarlane stabilizer actuators per the applicable Cessna Model Service Manual.

Refer to the applicable Cessna Model Service Manual for installation/maintenance instructions and the applicable Cessna Illustrated Parts Catalog for component part numbers.

NOTICE

Weight and balance or operating limitation changes shall be entered in the appropriate aircraft record. An alteration must be compatible with all previous alterations to assure continued conformity with the applicable airworthiness requirements.

8. Description of Work Accomplished
(If more space is required, attach additional sheets. Identify with aircraft nationality and registration mark and date work completed.)

N9248C	10/12/2024
Nationality and Registration Mark	Date

Installed McFarlane Aviation, Inc stabilizer Jack-Screw actuators in accordance Supplemental Type Certificate SA02112AK. Installation was accomplished in accordance with North Pacific Welding drawing number C-0712500, dated March 31, 2000 and the Cessna Skywagon 180 and 185 series service manual or McFarlane Aviation, Inc Drawing 1111-1, revision original dated June 1, 2010 or later FAA approved revisions.

Instructions for Continued airworthiness to be accomplished in accordance with the manufacturer's instructions for Continued Airworthiness as presented in document 1111-1, McFarlane drawing 6465.

Weight and Balance change negligible

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☐ Additional Sheets Are Attached